



Aluminum Air Powered Vehicle

Project Report

Machine Elements Design (MDP311)

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Table of Contents

Table of Figures	3
Abstract	4
Introduction	4
Background	4
Project Objectives	4
Design Constraints	4
Propulsion Method Comparison	5
Comparison	5
Conclusion	5
System Overview	6
General Description	6
System Block Diagram	6
Nozzle Design	7
Design Objectives	7
Assumptions for Nozzle Design	7
Nozzle Type Selection	8
1. Thrust Requirement and Nozzle Diameter Determination	8
2. Thrust Produced by a Converging Nozzle (Choked Flow)	9
3. Throat Area and Diameter Calculation	9
Experimental Comparison of Throat Diameters	10
Mechanical Design	11
Pressure Tank Design and Analysis	11
Assumptions:	11
Hoop stress:	12
Shaft Analysis	13
Base Analysis	13
Initial Tension on Bolts	14
Braking Concept Description	14
Full Cad Design	15
Construction Drawing	16
Working Drawings	17

Car Physical Properties.....	21
Materials Breakdown	21
Stress Analysis	22
Boundary & Load Conditions	22
Structural Evaluation	22
Simulation	23
Conclusion	25

Table of Figures

Figure 1 Tank	11
Figure 1 Tank	11
Figure 2 Shaft loading diagrams.....	13
Figure 2 Shaft loading diagrams.....	13
Figure 3 Full Cad Design	15
Figure 3 Full Cad Design	15
Figure 4 assembly drawing.....	16
Figure 4 assembly drawing.....	16
Figure 5 Tank WD	17
Figure 5 Tank WD	17
Figure 6 Base	18
Figure 6 Base	18
Figure 7 Front Shaft with Braking.....	19
Figure 7 Front Shaft with Braking.....	19
Figure 8 rear shaft	20

Abstract

This report presents the design, manufacturing, and testing of an air-powered vehicle driven solely by compressed air energy. The vehicle utilizes a nozzle-based propulsion system to convert the stored pressure energy in a 3-liter, 10-bar air tank into kinetic energy that generates thrust. The design process includes fluid, mechanical, and electrical subsystems integration, with a focus on maximizing performance, minimizing weight, and ensuring reliable control and braking. Analytical calculations, CAD modeling, manufacturing procedures, and experimental testing were conducted to validate the design. The final prototype successfully completed the required 10-meter track and applied effective braking, demonstrating the feasibility and efficiency of the proposed design.

Introduction

Background

Air-powered vehicles represent a simple and safe approach to demonstrate energy conversion principles without using combustion or electric propulsion. Compressed air stores mechanical energy in the form of pressure, which can be efficiently converted into kinetic energy using fluid devices such as nozzles or turbines. This project aims to utilize these principles to design a competitive and lightweight vehicle.

Project Objectives

- Design and manufacture an air-powered vehicle that travels 10 meters.
- Maximize thrust efficiency using a nozzle-based propulsion system.
- Minimize total vehicle mass.
- Design a reliable braking system.
- Implement wireless control for valve actuation and braking.
- Document the complete engineering design process.

Design Constraints

- Maximum tank volume: 3 liters.
- Maximum pressure: 10 bar.
- Only compressed air used for propulsion.
- No electric motors allowed for propulsion.
- Steel structure except wheels, brakes, and aero kit.
- Wireless control required.

Propulsion Method Comparison

Comparison

Aspect	Nozzle	Turbine
Principle	Direct conversion of pressure to jet momentum	Converts air energy to rotational power
Complexity	Very simple (no moving parts)	Complex (blades, shaft, bearings)
Weight	Very light	Heavy
Manufacturing	Easy	Difficult and expensive
Losses	Low	High (mechanical + leakage)
Reliability	Very high	Moderate
Control	Fast response	Slower response
Suitability for project	5/5	2/5

Table 1 Nozzle vs Turbine

Aspect	Nozzle	Air Motor
Output	Direct thrust	Rotational torque
Mechanical losses	Very low	High (gears, friction)
Efficiency at short runs	High	Moderate
Control complexity	Simple (valve control)	Complex (speed control)
Maintenance	Minimal	High
Cost	Low	High
Weight	Light	Heavy
Response time	Instant	Delayed

Table 2 Nozzle vs Air Motor

Conclusion

The nozzle was selected due to its simplicity, lower weight, and higher reliability compared to a turbine, making it more suitable for a short-distance high-acceleration application. Although air motors provide controlled torque, the nozzle-based system offers faster response, lower losses, and better performance for short-distance sprint applications such as the 10-meter track.

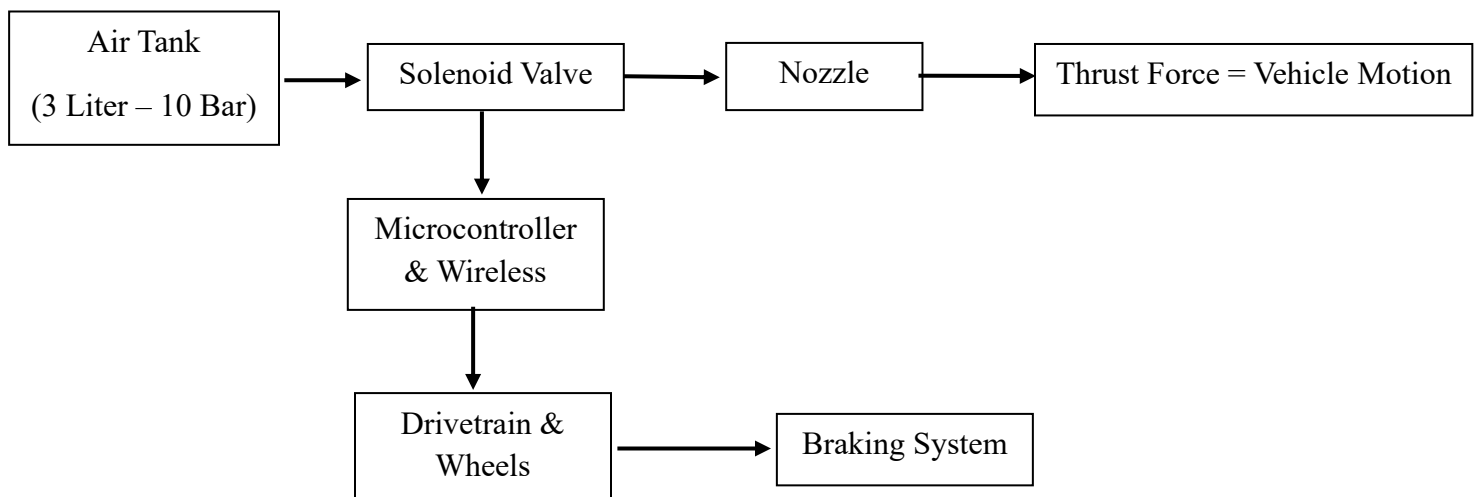
System Overview

General Description

The air-powered vehicle consists of integrated subsystems that convert compressed air energy into vehicle motion. Compressed air is stored in a high-pressure tank and released through a solenoid valve into a converging nozzle, where pressure energy is converted into kinetic energy to generate thrust. The generated thrust accelerates the vehicle along the track.

A motorized braking system is used to stop the vehicle after the finish line by applying a controlled braking torque to the rear axle. All propulsion and braking actions are controlled wirelessly using a microcontroller-based system, ensuring safe and repeatable operation.

System Block Diagram



Nozzle Design

Design Objectives

- Maximize exit velocity.
- Minimize losses.
- Stable flow.
- Easy manufacturing.
- Compatible with available pressure.

Assumptions for Nozzle Design

The following assumptions were made to simplify the analysis and design of the nozzle:

1. **Steady Flow Assumption**

The airflow through the nozzle is assumed to be steady, meaning that flow properties do not change with time.

2. **Incompressible Flow**

Since the air velocity is relatively low, the flow is assumed to be incompressible and density remains constant.

3. **One-Dimensional Flow**

The flow is assumed to be one-dimensional along the nozzle axis, and variations in the radial direction are neglected.

4. **Negligible Heat Transfer**

Heat transfer between the air and nozzle walls is neglected, and the process is considered adiabatic.

5. **No Friction Losses**

The effect of wall friction is neglected, and the nozzle is assumed to be smooth with no energy losses.

6. **Uniform Inlet Conditions**

The velocity and pressure of air at the nozzle inlet are assumed to be uniform.

7. **Ambient Exit Pressure**

The air at the nozzle exit is assumed to discharge to atmospheric pressure.

8. **Rigid Nozzle Material**

The nozzle is assumed to be rigid with no deformation under operating pressure.

9. **No Leakage**

There is no air leakage between connections or through the nozzle walls.

10. **Ideal Gas Behavior**

Air is assumed to behave as an ideal gas throughout the nozzle.

Nozzle Type Selection

A convergent nozzle was selected for this air-powered vehicle due to the operating conditions of the system. The working fluid is compressed air, and the pressure ratio is not high enough to justify the use of a convergent–divergent nozzle. Under these conditions, the flow accelerates effectively to its maximum velocity at the nozzle exit while remaining subsonic.

The convergent nozzle offers several advantages, including simpler geometry, lower manufacturing cost, reduced weight, and sufficient thrust generation for the intended application. Therefore, it represents an efficient and reliable choice that satisfies the performance requirements of the system without unnecessary design complexity.

1. Thrust Requirement and Nozzle Diameter Determination

1. Required Thrust for Vehicle Motion

To achieve noticeable acceleration over a short 10 m track, a target acceleration of about:

Assuming:

- Initial velocity $u = 0$
- Constant acceleration a
- Distance $s = 10\text{m}$
- Time to finish track $t \approx 3.5\text{ s}$ (typical competitive target)

The kinematic equation is:

$$s = \frac{1}{2}at^2$$

Solving for acceleration:

$$a = \frac{2s}{t^2}$$

$$a = \frac{2 \times 10}{(3.5)^2} = \frac{20}{12.25} \approx 1.63 \text{ m/s}^2$$

$$a \approx 1.5 \text{ m/s}^2$$

Rolling resistance is estimated as:

Assume $C_{rr} = 0.02$

$$F_{rr} = C_{rr}mg$$

$$F_{rr} = 0.02 \times 4 \times 9.81 = 0.78 \text{ N}$$

The required thrust is therefore:

$$F = ma + F_{rr}$$

$$F = (4)(1.5) + 0.78 = 6.78 \text{ N}$$

$$F_{required} \approx 7 \text{ N}$$

2. Thrust Produced by a Converging Nozzle (Choked Flow)

Given:

- Tank pressure $P = 8 \text{ bar} = 8 \times 10^5 \text{ Pa}$
- Atmospheric pressure $P_a = 1 \text{ bar}$
- Air ($\gamma = 1.4$, $R = 287$)
- Nozzle efficiency = 100%
- $T = 300 \text{ K}$

Since:

$$\frac{P_0}{P_a} = 8 > 1.89$$

the flow is choked, and the exit velocity is:

$$V_e = \sqrt{\gamma R T_0 \left(\frac{2}{\gamma + 1} \right)}$$
$$V_e \approx 315 \text{ m/s}$$

Thrust is:

$$F = \dot{m} V_e$$

So the required mass flow rate is:

$$\dot{m} = \frac{F}{V_e} = \frac{7}{315} = 0.022 \text{ kg/s}$$

3. Throat Area and Diameter Calculation

For choked flow:

$$\dot{m} = C_d A_t P_0 \sqrt{\frac{\gamma}{R T_0} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{\gamma-1}}}$$

Using:

- $C_d = 0.9$
- $P_0 = 8 \times 10^5$
- $T_0 = 300 \text{ K}$

This gives:

$$A_t \approx 1.25 \times 10^{-5} \text{ m}^2$$

Throat diameter:

$$d_t = \sqrt{\frac{4 A_t}{\pi}} = 0.0040 \text{ m}$$

$$\boxed{d_t \approx 4 \text{ mm}}$$

Experimental Comparison of Throat Diameters

Table 3 Experimental Results

Throat Diameter (mm)	Estimated Thrust (N)	Time (s)	Finished 10 m?	Observations
4 mm	~2.8 N	3.6 s	Yes	Strong initial acceleration, smooth motion
6 mm	~4.5 N (short burst)	4.5 s	Yes	Pressure dropped quickly, thrust decayed
8 mm	~6.0 N (very short burst)	>6 s	No	Tank emptied too fast, weak mid-run acceleration
10 mm	~7.5 N (instant peak)	—	No	Immediate pressure loss, vehicle stalled

Mechanical Design

Pressure Tank Design and Analysis

Assumptions:

- Working fluid: Compressed air
- Maximum internal pressure: $P = 10 \text{ bar} = 1.0 \times 10^6 \text{ Pa}$
- Material: Aluminum
- Yield strength: $\sigma_y = 275 \text{ MPa}$
- Factor of safety: $n=2$
- Allowable stress: $\sigma_{allow} = \frac{\sigma_y}{n} = 137.5 \text{ MPa}$
- Tank radius: $r=70 \text{ mm}$

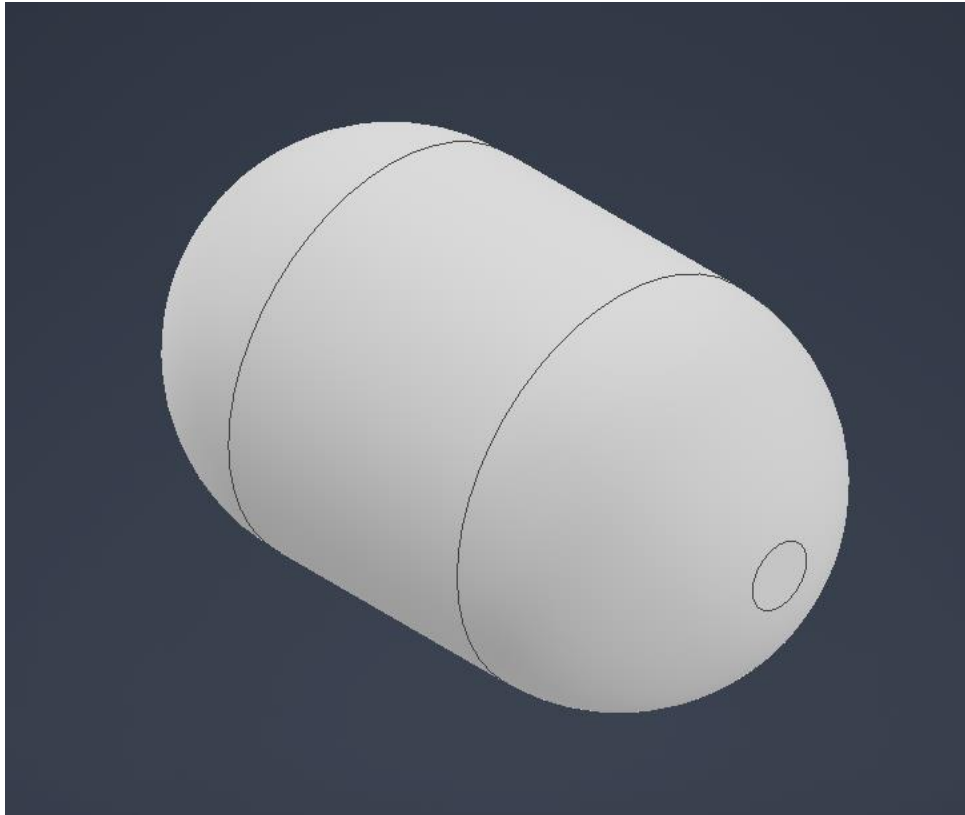


Figure 1 Tank

Figure 2 Tank

Hoop stress:

$$\sigma_h = \frac{Pr}{t}$$

Where:

- σ_h = hoop stress (Pa)
- P = internal pressure (Pa)
- r = internal radius (m)
- t = wall thickness (m)

Rearranging the equation:

$$t = \frac{Pr}{\sigma_{allow}}$$

$$t = \frac{(1.0 \times 10^6)(0.07)}{107.5 \times 10^6}$$

$$t = 6.51 \times 10^{-4} \text{ m}$$

$$t = 0.65 \text{ mm}$$

Selected Thickness:

$$t = 1.0 \text{ mm}$$

Design Validation:

$$\sigma_h = \frac{Pr}{t} = \frac{1.0 \times 10^6 \times 0.07}{0.001} = 70 \text{ MPa}$$

$$70 \text{ MPa} < 107.5 \text{ MPa}$$

The tank is safe under maximum operating pressure

Shaft Analysis

Assume:

Thrust force = 6 N

Mass = 3kg

The car is symmetric

Load on each wheel = 7.3575 N

Coefficient of friction = 0.2

Using:

$$\tau_{max} = \frac{16}{\pi d^3} \sqrt{(K_b M)^2 + (K_t T)^2}$$

$$\tau_{allow} = \frac{16}{\pi d^3} \times 0.662$$

$$d = \sqrt[3]{6.28 \times 10^{-8}}$$

$$d = 0.00397 \text{ m} = 3.97 \text{ mm}$$

$$d^3 = \frac{16 \times 0.662}{\pi \times 53.75 \times 10^6}$$

$$d^3 = 6.28 \times 10^{-8}$$

$$d_{min} \approx 4 \text{ mm}$$

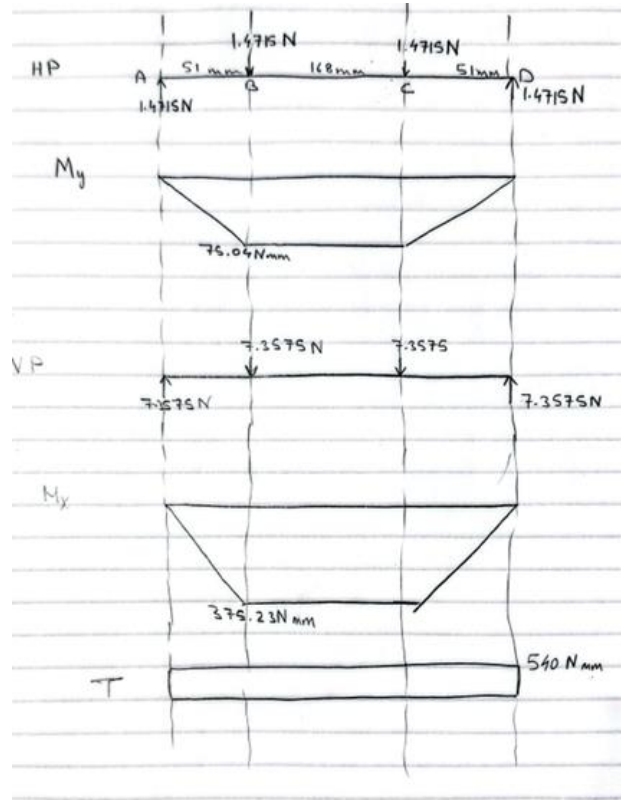


Figure 3 Shaft loading diagrams

Base Analysis

Assume:

- Vehicle mass= 3 kg
- Weight= $3 \times 9.81 = 29.4 \text{ N}$
- Base length between wheel supports:
 $L=168 \text{ mm}$
- Base width:
 $b=296 \text{ mm}$
- Material: Stainless steel sheet
Yield strength:
 $\sigma_y=215 \text{ MPa}$

Maximum Bending Moment:

$$M_{max} = \frac{WL}{4}$$

$$M_{max} = \frac{29.4 \times 0.168}{4}$$

$$M_{max} = 1.235 \text{ Nm}$$

Bending Stress:

$$\sigma = \frac{6M}{bt^2}$$

$$\sigma = \sigma_{allow}$$

$$53.75 \times 10^6 = \frac{6 \times 1.235}{0.296 \times t^2}$$

$$t^2 = \frac{7.41}{15.9 \times 10^6}$$

$$t^2 = 4.66 \times 10^{-7}$$

$$t = 0.000683 \text{ m}$$

$$t_{min} = 0.68 \text{ mm}$$

Figure 4 Shaft loading diagrams

Initial Tension on Bolts

$$\sigma = F_i / A_c = F_i / 11.885$$

$$\tau = T_r / J = 0.087 T_i$$

$$T_i = F_i * (d_m / 2) * ((\tan \alpha + \mu) / (1 - \mu * \tan \alpha)) = 0.6716 F_i$$

$$\tau = 0.058 F_i$$

$$\tau_{\max} = 80 \text{ MPa}$$

$$\tau_{\max} = \sqrt{\left(\frac{\varepsilon}{2}\right)^2 + \tau^2}$$

$$\text{Then } F_i = 1116.5 \text{ N}$$

$$\text{And there are 8 each have } F_{ex} = 0$$

Braking Mechanism Design and Analysis

Braking Concept Description

The braking system is designed to stop the rotating shaft using a friction-based disk brake actuated by a linear solenoid.

A circular steel disk is rigidly mounted on the shaft. When braking is required, a linear solenoid is energized, pushing a friction pad against the disk surface, generating friction torque that opposes shaft rotation and brings it to rest.

- This mechanism was selected because it is:
- Simple and lightweight
- Fast in response (electromagnetic actuation)
- Easy to control electrically
- Reliable for low-speed, low-torque applications

The braking action is normally-off, meaning the solenoid only consumes power during braking, which improves energy efficiency.

Full Cad Design

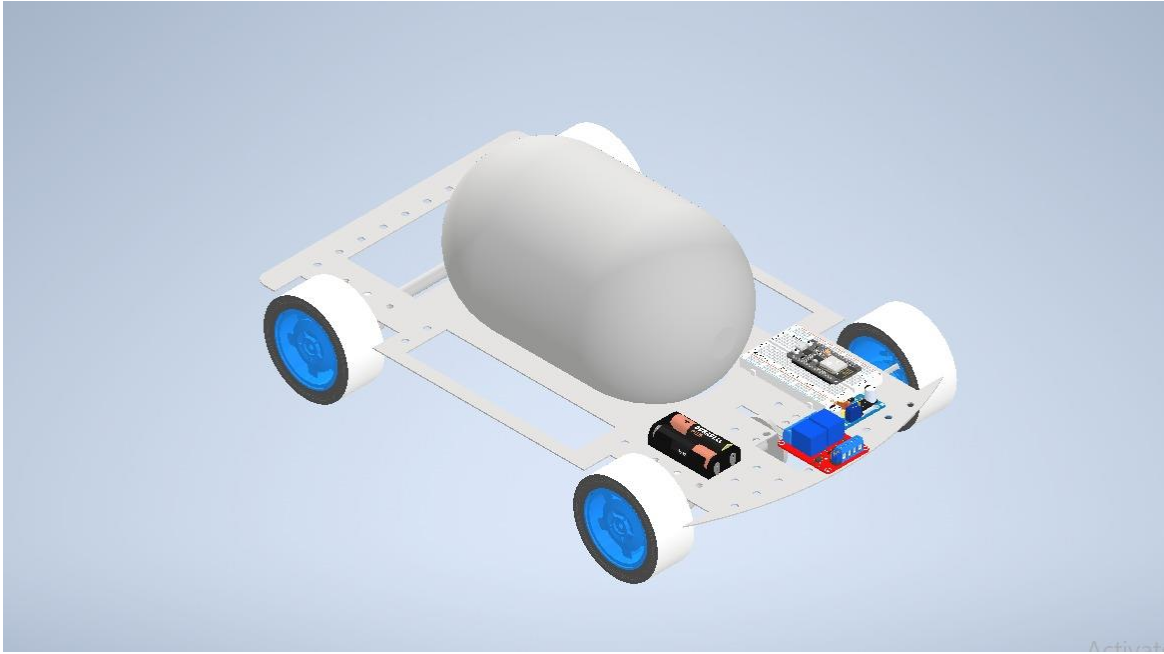


Figure 5 Full Cad Design

Figure 6 Full Cad Design

Construction Drawing

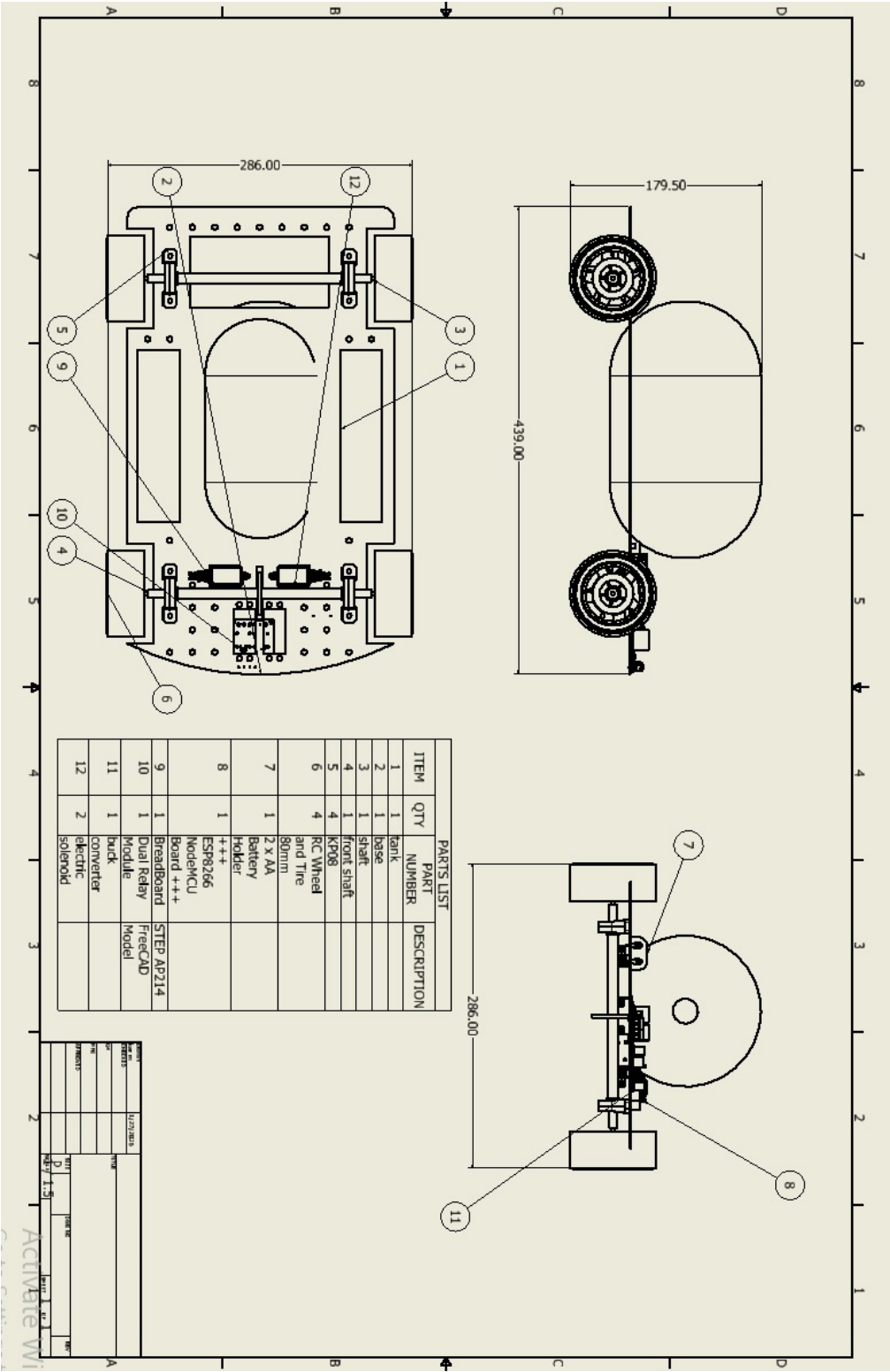


Figure 7 assembly drawing

Figure 8 assembly drawing

Working Drawings

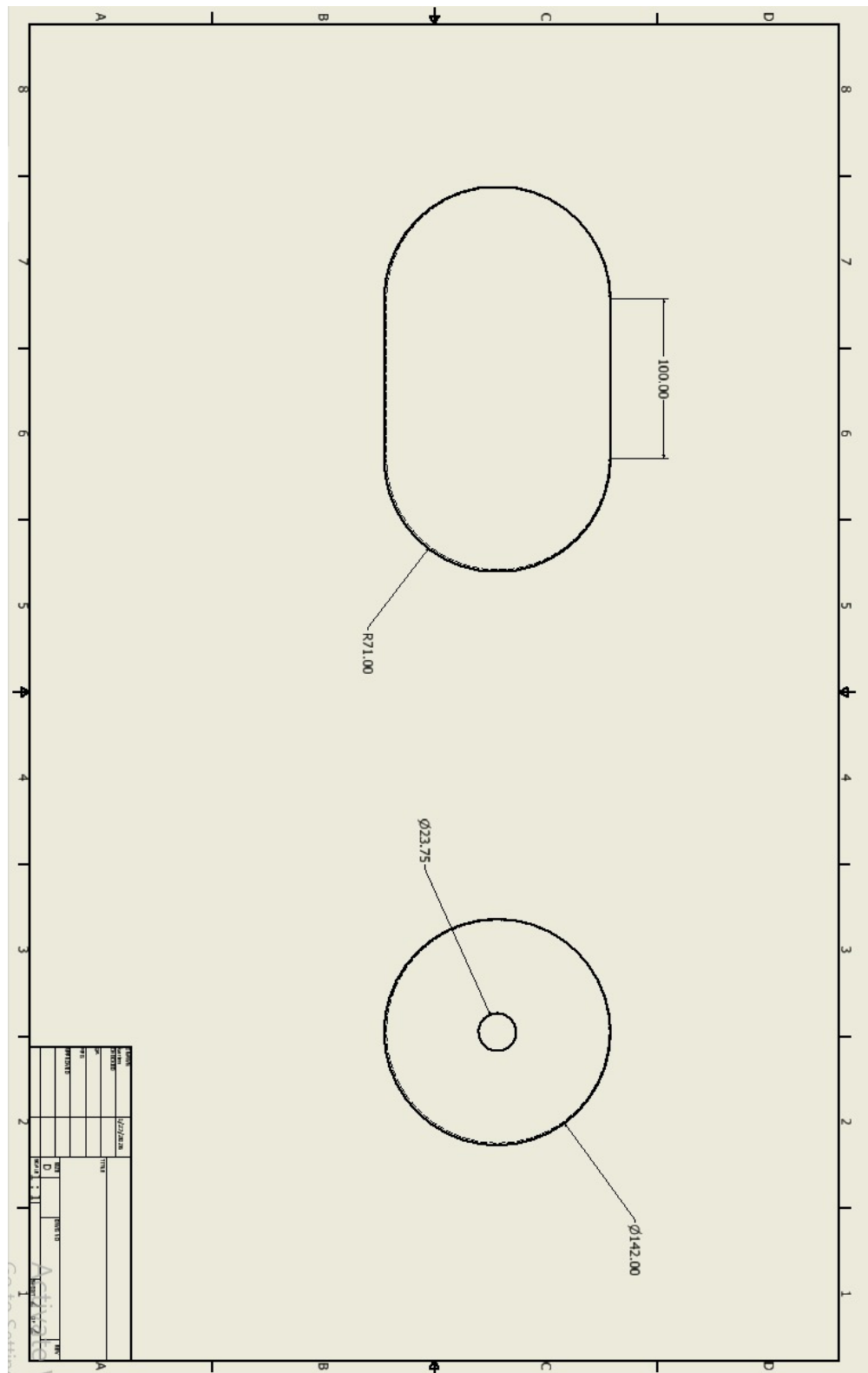


Figure 9 Tank WD

Figure 10 Tank WD

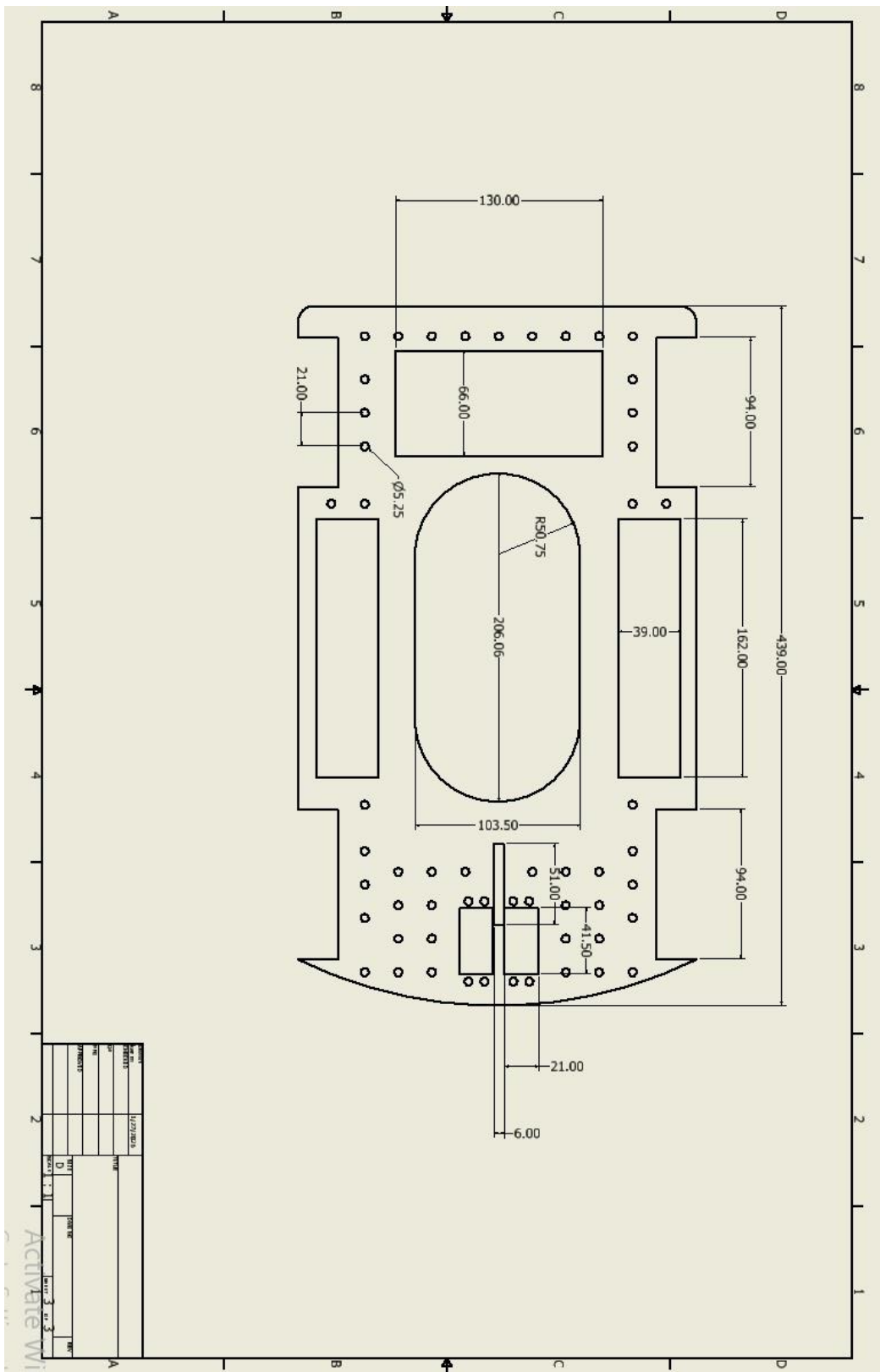


Figure 11 Base

Figure 12 Base

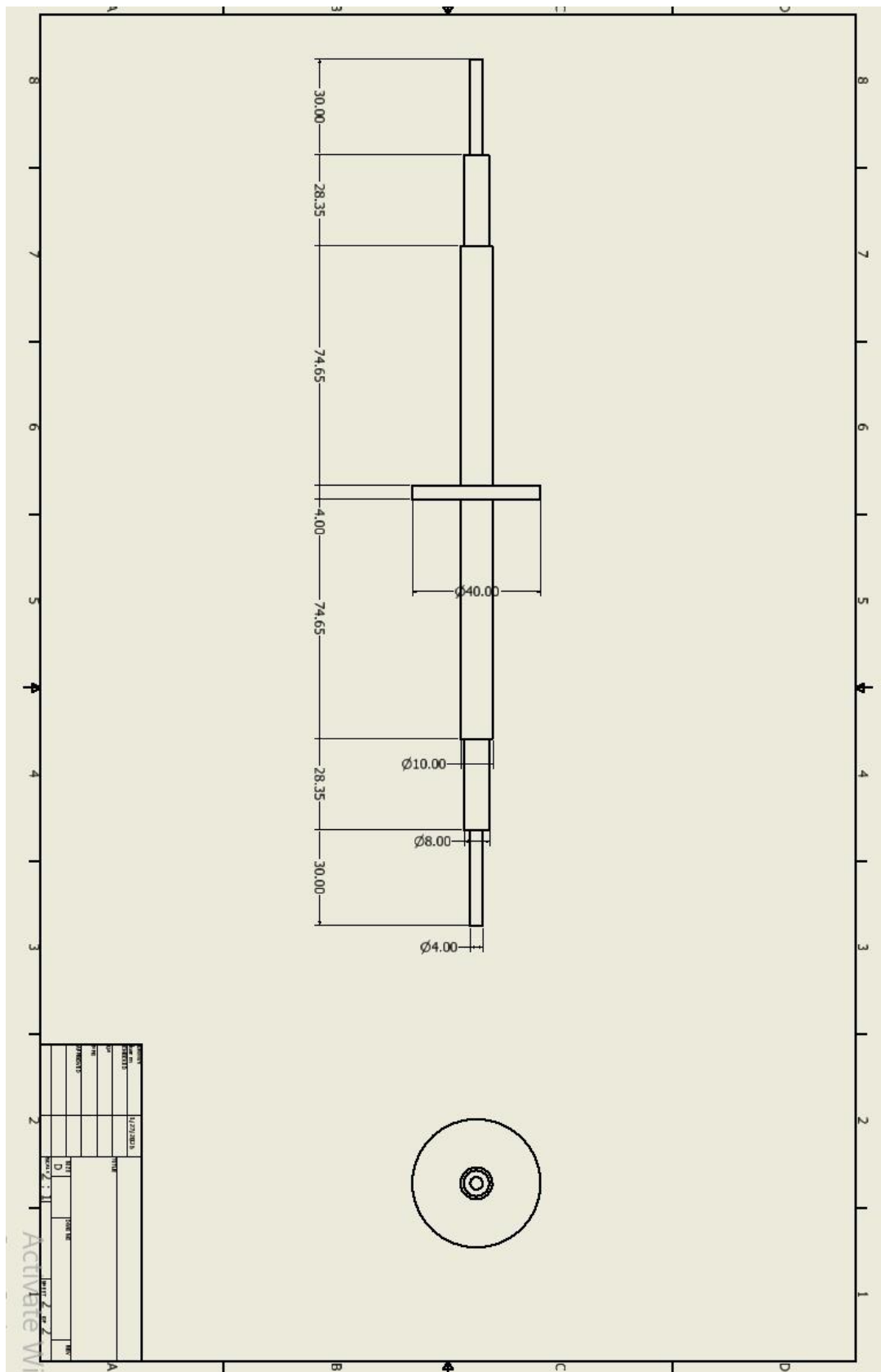


Figure 13 Front Shaft with Braking

Figure 14 Front Shaft with Braking

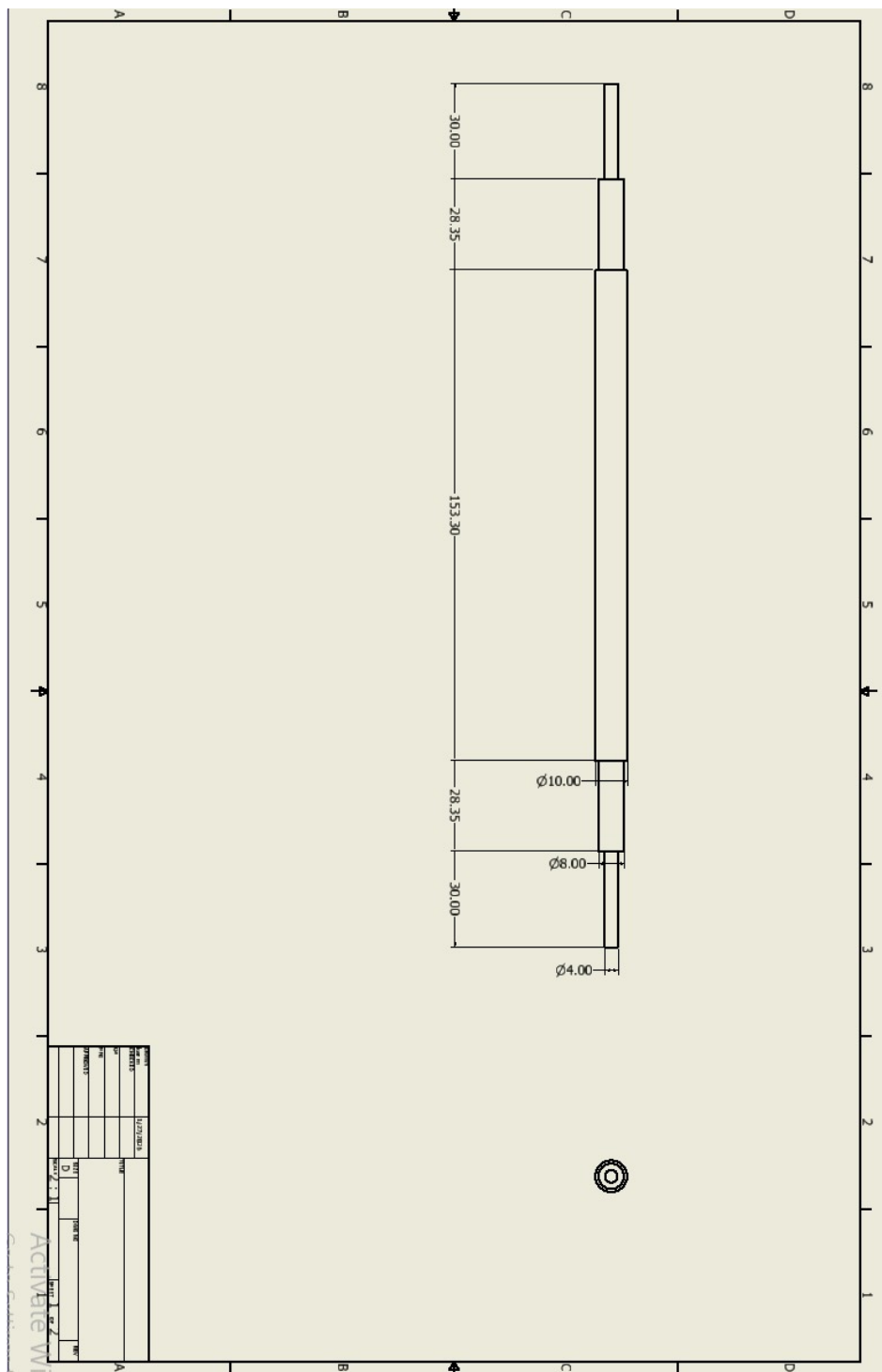


Figure 15 rear shaft

Car Physical Properties

The screenshot shows the 'iProperties: Assembly1' dialog box with the 'Physical' tab selected. The 'Material' section is empty. The 'Density' is set to 2.208 g/cm³ and 'Requested Accuracy' is Low. The 'General Properties' section includes checkboxes for 'Include Cosmetic Welds' and 'Include QTY Overrides', both of which are unchecked. The 'Center of Gravity' section displays mass and area values for X, Y, and Z axes. The 'Inertial Properties' section has three tabs: 'Principal', 'Global', and 'Center of Gravity', with 'Center of Gravity' currently selected. It shows mass moments of inertia (Ixx, Iyy, Izz) and products of inertia (Ixy, Iyz, Ixz) calculated using a negative integral.

Property	Value
Material	
Density	2.208 g/cm ³
Requested Accuracy	Low
Mass	1.087 kg (Relative E)
Area	632920.651 mm ²
Volume	492333.596 mm ³
Center of Gravity X	147.527 mm (Relative)
Center of Gravity Y	30.621 mm (Relative)
Center of Gravity Z	51.598 mm (Relative)
Mass Moments Ixx	8291.921 kg mm ²
Mass Moments Iyy	24897.157 kg mm ²
Mass Moments Izz	19155.071 kg mm ²
Products of Inertia Ixy	93.810 kg mm ³
Products of Inertia Iyz	-32.199 kg mm ³
Products of Inertia Ixz	2.430 kg mm ³

Materials Breakdown

- Frame Base, Tank Shell, Axle Shafts, Bearing Pillars: Aluminum 6061
- Tires & High-Friction Elements: Silicone Rubber

Stress Analysis

To validate the structural integrity of the lightweight aluminum chassis design under operating environments, a linear static stress analysis was performed within the Autodesk Inventor Simulation Environment.

Boundary & Load Conditions

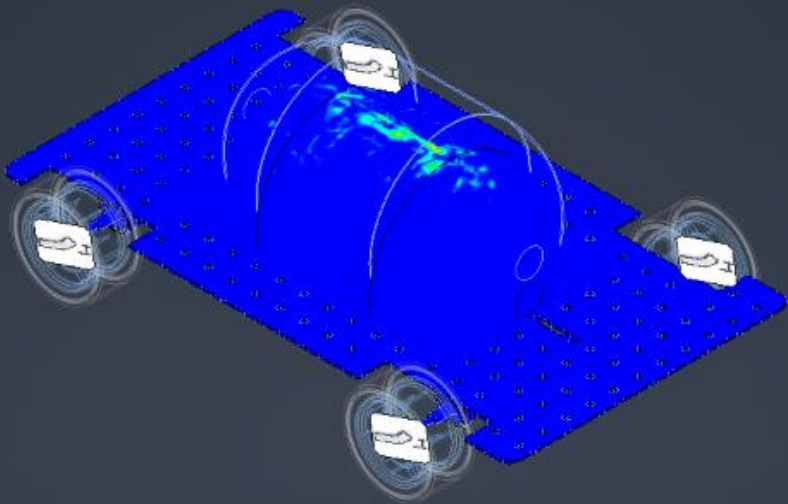
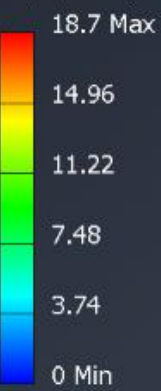
- Constraints: Fixed boundary constraints were applied to the four outer cylindrical mounting journals of the drive axles to simulate the ground-support reaction through the wheel assembly.
- Loads: A global gravitational acceleration load of 9.81 was applied vertically downwards to evaluate structural bending under the weight of the frame and storage tank.

Structural Evaluation

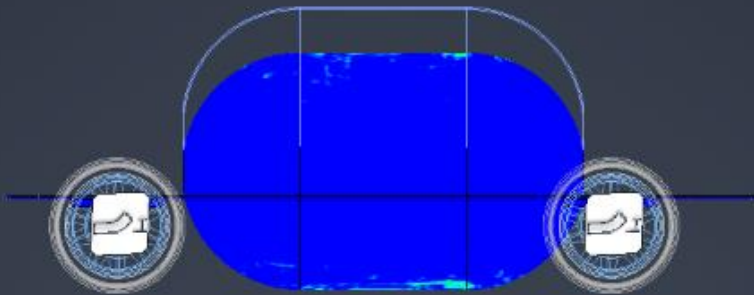
The simulation results indicate that the maximum induced internal stresses 7.328 MPa occur locally at the bonded interfaces where the thin-walled tank support brackets connect to the flat base plate. Because this peak value is well below the material's elastic yield limit 2.7, the vehicle operates entirely within a safe elastic window, with zero risk of permanent plastic deformation or structural failure.

Simulation

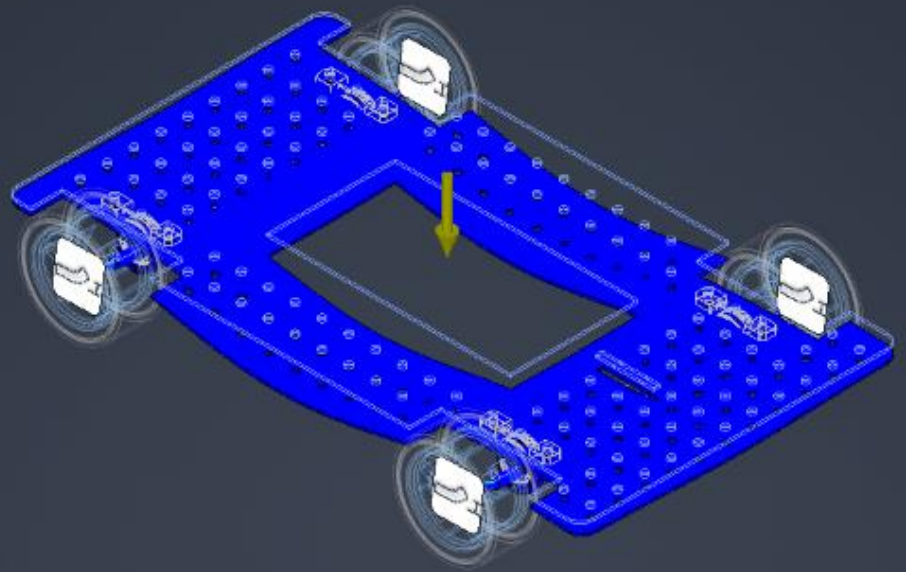
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Unit: MPa
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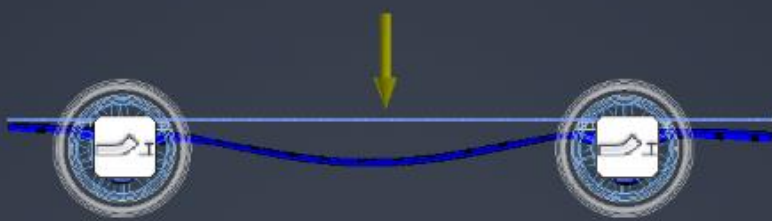
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Unit: MPa
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Unit: MPa
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Unit: MPa
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Conclusion

This project successfully achieved the comprehensive design, structural verification, and optimization of a high-efficiency, air-powered vehicle for short-track sprint performance. Through extensive propulsion system evaluations, a converging fluid nozzle was chosen over rotational turbine layouts due to its low mechanical friction losses, instant actuation characteristics, and minimal mass penalties. Analytical fluid calculations verified that an optimized 4 mm throat diameter establishes a choked flow condition under a 10 bar storage limit, providing a sustained 2.8 N initial thrust vector capable of covering the target 10-meter track layout within a highly competitive 3.6-second time domain without stalling. Transitioning the structural architecture from a traditional steel layout to a thin-walled Aluminum 6061 design reduced the vehicle's dead weight down to a highly efficient 1.087 kg. Linear static finite element analysis verified that the chassis remains structurally robust, with a peak localized Von Mises stress of just 7.328 MPa yielding a conservative factor of safety exceeding 37 times the material's yield limit. Paired with a fast-acting electromagnetic disk braking system, the integrated prototype proves the viability of using compressed green energy for high-acceleration, zero-emission mobile applications.